

DANIELE ANGELINI

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CURRENT POSITION

Research Fellow

MEMOTEF Department, Sapienza University of Rome. SSD: STAT/04-A

November 2025 - Present

Contract Professor - Matematica Finanziaria (6 CFU)

Scienze Aziendali Bachelor's course

Faculty of Economics, Sapienza, University of Rome. SSD: STAT/04-A

March 2026 - Present

Contract Professor - Matematica Finanziaria (3 CFU)

Scienze Aziendali Bachelor's course

Faculty of Economics, Sapienza, University of Rome. SSD: STAT/04-A

March 2026 - Present

Contract Professor - Mathematical Methods for Finance (9 CFU)

Quantitative Finance and Data Science for Economics Master's course

Department of Economics, University of Perugia. SSD: STAT/04-A

October 2025 - Present

Teaching Assistant - Mathematics

Faculty of Economics, Unitelma Sapienza

September 2025 - Present

Teaching Assistant - Applied Statistics and Econometrics

Luiss Guido Carli. SSD: ECON/05-A

Present

EDUCATION & TRAINING

Ph.D. in Mathematics for economic-financial applications

Sapienza University of Rome (Italy)

February 19, 2026

November 2022 - October 2025

Supervisor: Prof. Sergio Bianchi

Doctor Europaeus - Cum Laude

Thesis: Fair Volatility in a Multifractional World: A New Equilibrium Benchmark for Financial Markets

Escuela Internacional de Doctorado, University of Almería, Almería (Spain)

Visiting Research - Project research: Dynamic Heterogeneities in Complex Systems

Supervisor: Prof. Juan E. Trinidad Segovia

October 2025

Collegio Internazionale Ca' Foscari, Venezia (Italy)

Doctoral Colloquium on Risk Analytics

Session 2: New challenges on long-run risks

July - August 2025

Centre de Recerca Matemàtica, Barcelona (Spain)

5th Barcelona Summer School of Stochastic Analysis and Quantitative Finance

Courses: Rough volatility, signatures in stochastic finance, weather derivatives

July 2025

New York University, Tandon School of Engineering, New York City (USA)

Advanced Risk of Portfolio Management, Quant Bootcamp

July 2025

Ecole Supérieure d'Ingénieurs Léonard de Vinci, ESILV, Paris (France)

Visiting Research - Finance Group

Supervisor: Prof. Matthieu Garcin

January - July 2024

Academia de Studii Economice din București, Bucarest (Romania)

Blinded Mobility FOrReSIGHT Erasmus+ project: Artificial Intelligence

May 2023

Sapienza University of Rome, Rome (Italy)

Master Degree in Theoretical Physics

May 2019 - July 2022

109/110

Thesis: Financial market efficiency analysis with multifractional processes with random exponent and damped forced harmonic oscillator
July 20, 2022

ACADEMIC BACKGROUND

Teaching Assistant - Financial Mathematics *October - January 2026*
Faculty of Economics, Sapienza University of Rome

Teaching Assistant - Financial Mathematics *May - June 2025*
Faculty of Economics, Sapienza University of Rome

Teaching Assistant - Mathematics *May - June 2025*
Faculty of Economics, Sapienza University of Rome

Teaching Assistant - Black-Scholes formula & Fractional Analysis *December 2024*
Sapienza's School for Advanced Studies, SSAS

Teaching Assistant - Financial Mathematics *October - December 2024*
Faculty of Economics, Sapienza University of Rome

Teaching Assistant - Mathematics (Economics and Finance) *September - December 2024*
Faculty of Economics, Sapienza University of Rome

Teaching Assistant - Mathematics (Business sciences) *October - December 2023*
Faculty of Economics, Sapienza University of Rome

Teaching Assistant - Mathematics (Economics and Finance) *October - December 2023*
Faculty of Economics, Sapienza University of Rome

Teaching Assistant - Calculus I *February - December 2023*
Department of Electronic Engineering, Sapienza University of Rome

Teaching Assistant - Calculus II *February - December 2023*
Department of Electronic Engineering, Sapienza University of Rome

PRIZES

2025 - *Best Paper Award* in the 53th EBES Conference (Madrid)
2025 - Finalist for *Best Junior Paper*, AMASES Conference
2025 - DISF Award, Pontifical University of the Holy Cross
2023 - *Best Paper Award* in the 45th EBES Conference (Budapest)

SCOLARSHIPS & FUNDINGS

2026 - Funding awarded for XIII Bachelier World Congress
2025 - Funding for the *Starting Research* call.
2025 - Awarded participation in DoCRA, Ca' Foscari University of Venice
2025 - Awarded participation in ARPM Quant Bootcamp, New York University
2025 - Awarded participation at H2CU College Italia (New York)
2024 - Funding for the *Starting Research* call.
2023 - Funding for the *Medium project* call. Group leader: S. Bianchi.
2023 - Funding for the *Starting Research* call.
2023 - FOReSIGHT scholarship - Erasmus+ project (Bucarest)
2022 - Scholarship Sapienza - Dottorato di ricerca

MAIN SCIENTIFIC PUBLICATIONS

Refereed Journal Articles

1. Bianchi S., & Angelini, D. (2026) When is Volatility Fair? Hölder Regularity and Financial Risk. *Applied Mathematics and Computation*, 531, 130223.
2. Di Sciorio, F., & Angelini, D. (2026). Integrating the implied regularity (H_t) into implied volatility models: A study on free arbitrage model. *Journal of Economic Analysis*, 5(2), 136.
3. Angelini, D., & Bianchi, S. (2025). Kolmogorov–Smirnov estimation of self-similarity in long-range dependent fractional processes. *Physica D: Nonlinear Phenomena*, 476, 134697.
4. Bianchi, S., Pianese, A., Frezza, M., & Angelini, D. (2025). A new tool to detect financial data scaling. *Frontiers in Applied Mathematics and Statistics*, 11, 1527750.
5. Bianchi, S., Angelini, D., Pianese, A., & Frezza, M. (2023). Rough volatility via the Lamperti transform. *Communications in Nonlinear Science and Numerical Simulation*, 127, 107582.
6. Angelini, D., & Bianchi, S. (2023). Nonlinear biases in the roughness of a Fractional Stochastic Regularity Model. *Chaos, Solitons & Fractals*, 172, 113550.

Refereed Chapter Books

7. Angelini, D., & Bianchi, S. (2026) An High-Frequency Perspective in Realized Volatility: RFSV vs HAR. In *Mathematical and Statistical Methods for Actuarial Sciences and Finance*. Cham: Springer Nature Switzerland.
8. Angelini, D. (2026). Time-Varying Self-Similarity to detect Creativity in Non-Sentient Systems. *Servicios de Publicaciones de la Universidad de Almería*.
9. Bianchi, S., & Angelini, D. (2025). Roughness in VIX index and in Realized Volatility: Rolling Window Estimation by Randomized Kolmogorov-Smirnov Distribution. In *New Perspectives in Mathematical and Statistical Methods for Actuarial Sciences and Finance* (pp. 1–13). Cham: Springer Nature Switzerland.
10. Bianchi, S., Angelini, D., Frezza, M., Palazzo, A. M., & Pianese, A. (2024). Fair Volatility in the Fractional Stochastic Regularity Model. In *Mathematical and Statistical Methods for Actuarial Sciences and Finance* (pp. 61-66). Cham: Springer Nature Switzerland.

Other Publications

11. Angelini, D. (2026). Fair Volatility in a Multifractional World: A New Equilibrium Benchmark for Financial Markets. PhD thesis. Sapienza, University of Rome. Available on IRIS.

Refereed Proceedings

12. Angelini D., Bianchi, S. When is Volatility Fair? Hölder Regularity and Financial Risk. BFS World Congress 2026 - Book of Abstract, pp. 107.
13. Angelini D. (2026). Time-Varying Self-Similarity to detect Creativity in Non-Sentient Systems. FIMBSS 2026 - Book of Abstract, pp. 5.
14. Angelini D., & Garcin M. (2025). Market information of the Fractional Stochastic Regularity Model. XLIX AMASES 2025 - Book of Abstracts, pp. 103.
15. Angelini D., & Bianchi S. (2025). Kolmogorov–Smirnov estimation of self-similarity in long-range dependent fractional processes. 5th Barcelona Summer School of Stochastic Analysis and Quantitative Finance - Contributed talks.
16. Angelini D. (2025). Integrating the implied regularity into implied volatility models: A study on free arbitrage model. 12th General AMaMeF Conference Booklet of Abstracts (pp.4).

17. Angelini D. (2025). Kolmogorov-Smirnov Estimation of Self-Similarity in Long-Range Dependent Fractional Processes. Fractals Conference, contributed talks (p.1).
18. Di Sciorio, F., & Angelini, D. (2025). Integrating the implied regularity into implied volatility models: A study on free arbitrage model. Published: 13 June 2025 by MDPI in The 1st International Online Conference on Risk and Financial Management.
19. Bianchi S., Angelini D., Frezza M., & Pianese A. (2024). Kolmogorov-Smirnov Distribution and Self-Similarity of fractional Brownian motion. Book of abstract SMSA 2024 (pp. 17-18).

Paper under review

20. Angelini, D. (2026). (In) Efficient Market States and Rough Volatility Detected via Grünwald-Letnikov Fractional Derivative. ArXiv preprint arXiv:2606.27932.
21. Bianchi, S., Angelini, D. (2026). Asymptotic Two-Sample Kolmogorov Distribution for Fractional Gaussian processes.
22. Frezza, M., Angelini, D., Bianchi, S., Pianese, A. (2026). Entropic Contractions and Stochastic Regularity in S&P500 Portfolio during Financial Crisis.
23. Angelini, D. (2026). Artificial Roughness in Volatility Models: A Critical Review of Hurst Exponent Estimation.
24. Bianchi S., & Angelini, D. (2025) Roughness Analysis of Realized Volatility and Volatility Index through Randomized Kolmogorov-Smirnov Distribution. ArXiv preprint arXiv:2509.20015.
25. Bianchi S., Angelini, D., Frezza, M., & Pianese, A. (2025). From fair price to fair volatility: Towards an Efficiency-Consistent definition of financial risk. ArXiv preprint arXiv:2508.11649.
26. Angelini, D., & Garcin, M. (2025). Market information of the fractional stochastic regularity model. ArXiv preprint arXiv:2409.07159.

INVITED SEMINARS

1. University of Cassino and Southern Lazio, May 27, 2026.
Title: “Price Dynamics in a Multifractional World: Equilibrium, Regularity, and Fairness”
2. Universidad de Almería, Spain, September 29 - October 3, 2025.
Title: “Rough Volatility in delampertized processes via the Kolmogorov-Smirnov test”.
3. De Vinci Research Center - Axis Sèminaire Axe 3, La Dèfense, Paris. February 29, 2024.
Title: “Market In-Efficiency”.
4. Quantitative Finance Workshop at ESILV University, La Dèfense, Paris. January 30, 2024.
Title: “A Fractional Stochastic Regularity Model and Rough Volatility via the Lamperti transform”.

WORKSHOPS AND SCIENTIFIC EVENTS ORGANIZED

1. PhD Progress Workshop 2025, MEMOTEF Department, PhD in Applied Mathematics and Statistics, October 13, 2025.
Role: **Organizer**.

INTERNATIONAL CONFERENCES

1. MASES Meeting - Sapienza, University of Rome, July 6-7, 2026.
Title: “Asymptotic Two-Sampled Kolmogorov Distribution for Fractional Gaussian Processes”
Speaker

2. XIII Bachelier World Congress - University of Bologna, June 29 - July 3, 2026.
Title: “When is Volatility Fair? Holder Regularity and Financial Risk” **Speaker**
3. First International Meeting on Behavioral Social Sciences - University of Almeria, April 23-24, 2026.
Title: “Time-Varying Self-Similarity to detect Creativity in Non-Sentient Systems” **Speaker**
4. 12th International Conference Mathematical and Statistical Methods for Actuarial sciences and Finance (MAF 2026) - Centre de Recerca Matemàtica, Universidad Autònoma de Barcelona, April 8-10, 2026.
Title: “Rough Volatility vs HAR: An High-Frequency Perspective” **Speaker**
5. XXVII Workshop on Quantitative Finance - University of Bergamo, March 30 - April 1, 2026.
Title: “When is Volatility Fair? Holder Regularity and Financial Risk” **Speaker**
6. 53rd EBES Conference - Budapest at the Mathias Corvinus collegium, October 16-18, 2025.
Title: “Fair Volatility: A Framework for Reconceptualizing Financial Risk”
7. AMASES XLIX annual conference. University of Florence, Florence, 11-13 September, 2025.
Plenary Session: Best Junior Paper.
Title: “Market information of the Fractional Stochastic Regularity Model” **Speaker**
8. Doctoral Colloquium on Risk Analytics - Session 2, International College Ca’ Foscari, Venice, August 6, 2025.
Title: “Self-similarity estimation using a Generalized Kolmogorov–Smirnov distribution” **Speaker**
9. Doctoral Colloquium on Risk Analytics - Session 2, International College Ca’ Foscari, Venice, July 31, 2025.
Title: “Generalized Kolmogorov–Smirnov distribution via a pseudo-fractional Brownian bridge” **Speaker**
10. 5th Barcelona summer school of Stochastic Analysis and Quantitative Finance - Centre de Recerca Matemàtica, Universidad Autònoma de Barcelona, July 21-25, 2025.
Title: “Kolmogorov–Smirnov estimation of self-similarity in long-range dependent fractional processes” **Speaker**
11. New perspectives in Mathematical and Statistical Methods for Actuarial Sciences and Finance, Waiting for MAF 2025 - University of Salerno, Salerno June 27-28, 2025. Poster session.
Title: “Roughness in VIX index and in Realized Volatility: Rolling Window Estimation by Randomized Kolmogorov-Smirnov Distribution” **Speaker**
12. 12th General AMaMeF Conference - University of Verona, departiment of Economics, June 23-27, 2025.
Title: “Integrating the implied regularity into implied volatility models: A study on free arbitrage model” **Speaker**
13. Cornell Conference on Analysis, Probability, and Mathematical Physics on Fractals (Fractals 8) - Cornell University, departiment of Mathematics, Ithaca (New York), June 15-20, 2025.
Title: “Kolmogorov-Smirnov Estimation of Self-Similarity in Long-Range Dependent Fractional Processes” **Speaker**
14. The 1st International Online Conference on Risk and Financial Management (IOCRF 2025), session Financial Innovations and Technology. SciForum, 17-18 June 2025.
Title: “Integrating the implied regularity into implied volatility models: A study on free arbitrage model”.
15. 11th International Conference Mathematical and Statistical Methods for Actuarial sciences and Finance (MAF 2024) - University of Le Havre Normandie, Le Havre Cedex, April 4-6,

2024.

Title: "Fair volatility in the Fractional Stochastic Regularity model"

16. 15th Workshop on Stochastic Models, Statistics and Their Applications (SMSA 2024) - TU Delft, Netherlands, March 13-15, 2024.

Title: "Kolmogorov-Smirnov Distribution and Self-Similarity of fractional Brownian motion"

17. 45th EBES Conference - Budapest at the Mathias Corvinus collegium, October 11-13, 2023.

Title: "Hurst-Hölder Regularity and Fair Volatility"

18. 5th edition Quantitative Finance & Financial Econometrics 2023 at the Aix-Marseille School of Economics (AMSE), June 8-9, 2023.

Title: "A Fractional Stochastic Regularity Model" **Speaker**

OTHER CONFERENCES

19. PhD Progress Workshop 2025, October 13, 2025.

Title: "A Kolmogorov-Smirnov Approach to Self-Similarity in Dependent Processes."

20. XV edition of "Giornate della Ricerca MEMOTEF", 29-30 May 2025, Rome.

Title: "Market information of the fractional stochastic regularity model".

21. XV Workshop SISRI - "Nulla di nuovo sotto il sole? Paradigmi e innovazione nelle scienze", 24-25 May 2025, Rome.

Title: "Time-varying self-similarity to detect creativity in non-sentient systems".

22. XIII edition of "Giornate della Ricerca MEMOTEF", 27-28 June 2023, Rome.

Title: "Fractional Stochastic Regularity Model".

SCHOLARLY REVIEWS

1. "Physica A: Statistical Mechanics and its Applications"

2. "IEEE Transactions on Biomedical Engineering"

3. "Frontiers in Physics"

4. "PeerJ Computer Science"

5. "Communications in Nonlinear Science and Numerical Simulation"

COMPUTER SKILLS AND LANGUAGE CERTIFICATES

Risk Analysis: ARPM Quantboot Camp certificate

Programming languages: Python, C, C++, R, MatLab, JAVA, JavaScript and Visual Basic

Machine Learning: Pandas, Numpy and Scikit-learn

Web programming languages: HTML and CSS

Database management: PostgreSQL, MongoDB, MS Access

Software & Tools: MS Office, Latex, RStudio, GNUplot

Language certificates: DELF B1